INDIVIDUAL PROJECT REPORT

Review 1 — Problem Identification & Proposal (24/8/2026)

AI-Driven Edge–Cloud Network for Cyclone Prediction and Resilient Ocean Telemetry

Student	Register No.	Team No.	My role
Anvay Sharma	24BCE0824	_____	Edge gateway and edge intelligence (classification, satellite image analysis, ML models)

Team: Laksh Ahuja (24BDS0212), Anvay Sharma (24BCE0824), Sreehari (24BDS0417). **Repository:** https://github.com/laksh-ahuja06/AI-DRIVEN-EDGE-CLOUD-NETWORK

Summary

Our project is a cyclone-monitoring network that must work over slow, lossy links. In this report I explain the problem from the point of view of the network edge: deciding *at the gateway* which data is urgent, and reducing heavy satellite images to the information that actually matters before they consume the backbone. I present the evidence, the literature I surveyed, our proposed solution and my plan.

1. Problem Statement

Edge devices in a cyclone-monitoring network forward everything they receive, in arrival order and at full size, so a multi-megabyte satellite image or a routine reading can delay the one packet that shows a storm is intensifying.

- **Why it matters:** the earlier a dangerous trend is recognised, the more time there is to warn and evacuate.
- **Who is affected:** fishing communities and coastal residents, and meteorological and disaster agencies that depend on near-real-time data.
- **If unsolved:** a threshold-only system gives zero warning before a threshold is crossed, and raw images can saturate the link for many minutes.

Objectives of My Part

Objective	Measurable success criterion
Classify every packet at the gateway as CRITICAL or NORMAL	No breach of 60 km/h / 990 hPa missed
Warn before thresholds are crossed using the trend of readings	Positive lead time in test scenarios
Cut satellite data sent over the backbone	> 95 % fewer bytes than the raw frame
Gateway that never crashes on bad input	Malformed packets counted and dropped
Fair, repeatable evaluation of every model	Held-out scenarios and a simple baseline for each

2. Evidence the Problem is Real

Table 1: Evidence that the problem is real

Source	Published fact	Why it matters for my part
India Meteorological Department [1]	India has a coastline of about 7,500 km; the 1999 Odisha super cyclone caused about 10,000 deaths.	The scale of exposure makes early warning valuable.
WMO Atlas 1970–2019 [2]	Tropical cyclones are among the leading causes of human and economic losses from weather, climate and water extremes.	Losses justify investing in faster, smarter edge decisions.
Engineering estimate (our calculation)	A 4-band 16-bit frame at 512 × 512 pixels is 2,097,152 bytes; lossless compression typically removes only a fraction of that for noisy imagery.	Sending images unprocessed is not viable; edge analysis must cut data volume by orders of magnitude.

3. Literature Survey

Table 2: Literature survey (15 articles)

#	Author / Year	Method used	Result	Limitation
1	Dvorak, 1975 [3]	Analyst reads cloud patterns in satellite images	Global operational standard	Subjective; needs experts
2	Velden et al., 2006 [4]	30-year review of the Dvorak technique	Documents accuracy and pitfalls	No automation for edge devices
3	Olander & Velden, 2007 [5]	Advanced Dvorak Technique (objective, IR)	Automated intensity estimates	Runs centrally on full images
4	Pradhan et al., 2018 [6]	Deep CNN on IR imagery	Learned intensity estimation	Needs GPU and large labelled data

#	Author / Year	Method used	Result	Limitation
5	Chen et al., 2019 [7]	CNN using multi-channel satellite data	Improved intensity estimates	Heavy model; central processing
6	Chen et al., 2020 [8]	Review of ML in cyclone forecasting	Maps ML methods to forecast tasks	No network or edge aspect
7	Denby & Lucia, 2020 [9]	Orbital edge computing on nanosatellites	Process imagery where it is captured	Space segment, not ground stations
8	Giuffrida et al., 2022 [10]	On-board DNN filters cloudy images (Φ-Sat-1)	Saves downlink by discarding useless data	Cloud screening only, not storm features
9	Hochreiter & Schmidhuber, 1997 [11]	LSTM recurrent network	Learns long time dependencies	Generic; needs a framework
10	Lim & Zohren, 2021 [12]	Survey of deep time-series forecasting	Compares RNN, attention, hybrids	Not edge-constrained
11	Chen & Guestrin, 2016 [13]	Gradient-boosted trees (XGBoost)	Strong tabular classifier	Less transparent than linear models
12	Shi et al., 2016 [14]	Edge computing vision	Motivates edge analytics	No algorithms
13	Mao et al., 2017 [15]	Mobile edge computing survey	Offloading decisions at the edge	Cellular focus
14	Shelby et al., 2014 [16]	CoAP protocol	Compact messages for constrained devices	No content-aware priority
15	Blake et al., 1998 [17]	DiffServ traffic classes	Class-based forwarding	Static classification

Gap: image-based cyclone analysis (Dvorak, ADT, CNNs) runs centrally on full images, and on-board satellite AI so far only screens out cloudy frames. Network classification uses fixed rules on current values. Nobody we found extracts storm features at the *ground station* and sends only those, or uses the *trend* of a sensor's readings at the gateway to raise priority before a threshold is crossed. These two ideas are my contribution to the project.

4. Proposed Solution Form

Form: coding and modelling. The edge gateway and the edge-AI functions are software by nature. We will first build a rule-based gateway (thresholds on wind and pressure) so the pipeline can be verified, then add lightweight models that can run on a laptop or a Raspberry Pi. A hardware-first approach was rejected because the decisions at the edge must be tested on many storm scenarios, which only simulation provides.

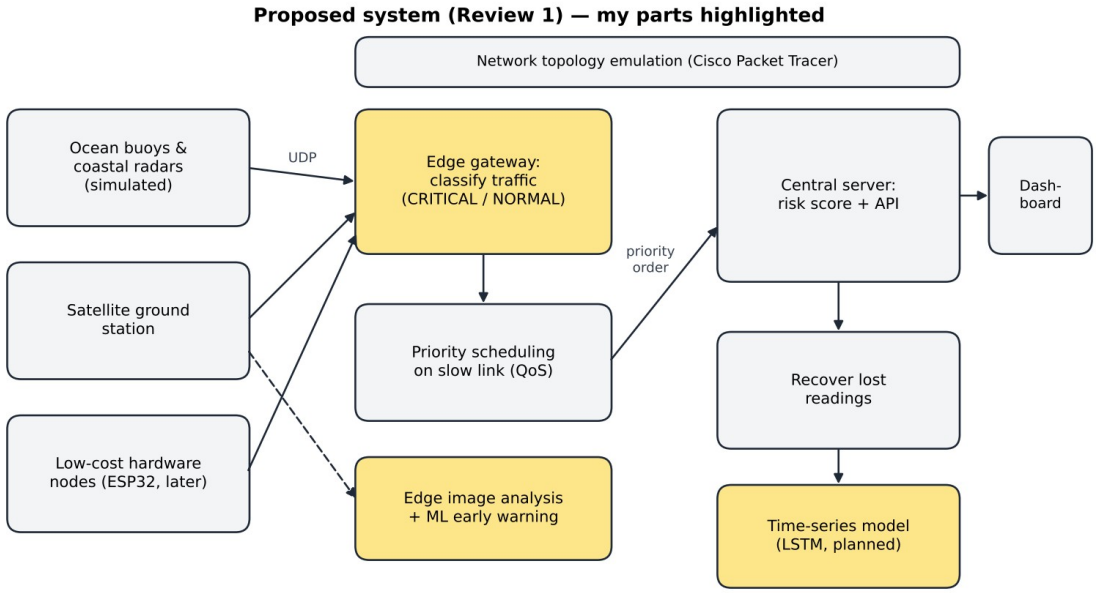


Figure 1: Proposed block diagram; my parts (gateway classification, edge image analysis, ML / LSTM) are highlighted

Planned Method for My Part

Component	Planned method
Gateway	Two threads — receiver and forwarder — joined by a priority queue; threshold rule first so the pipeline can be verified.
Early warning	Features from the last few readings (pressure falling, wind rising, distance to the thresholds) fed to a small, explainable classifier.
Satellite images	Extract the cues forecasters use — cold cloud tops, symmetry, an eye — at the ground station and send only those numbers.
Evaluation	Compare every model with a simple baseline on scenarios it has not seen.

5. Initial Technical Feasibility

Table 3: Feasibility of my part

Area	Status
Skills I have	Python, NumPy, basic machine learning (regression, classification), threading
Skills I must learn	Image processing on multi-band data, recurrent networks (LSTM) and how to train them, evaluating models fairly (held-out data, baselines)
Tools needed / access	Python, NumPy, Matplotlib (free); laptop CPU is enough for small models; real satellite imagery from public portals (MOSDAC) later
Risks	No labelled satellite data (mitigation: synthetic frames with known intensity); ML may raise false alarms (mitigation: keep the threshold rule as a safety floor); deep-learning frameworks may be too heavy for edge devices

Table 4: Risk register for my part

Risk	Likelihood	Impact	Mitigation
No labelled satellite images	High	High	Synthetic frames with known storm intensity
ML classifier raises false alarms	Medium	Medium	Keep the threshold rule as a safety floor
Deep-learning frameworks too heavy for the edge	Medium	Medium	Prefer small NumPy models
Gateway crashes on bad packets	Medium	High	Validate every packet before use

6. Team Role Split and Timeline to Review 2

Table 5: Team role split

Member	Role	Specific tasks (R1 → R3)
Laksh Ahuja (24BDS0212)	Central server, dashboard & network QoS	Prediction server and REST API, live dashboard, priority scheduling on the slow link, cyclone simulation model, demo launcher, automated testing
Anvay Sharma (24BCE0824)	Edge gateway & edge intelligence	Gateway classification and forwarding, satellite image analysis at the edge, ML early-warning classifier, LSTM implementation, evaluation and logs
Sreehari (24BDS0417)	Sensing layer, hardware & data recovery	Sensor-node and satellite-station simulators, packet-loss model, reconstruction of lost readings, model training data, ESP32 hardware nodes

My role is the intelligence at the edge: the gateway that classifies and forwards each packet, and the models that let it recognise danger early and shrink satellite data.

Table 6: My weekly plan from Review 1 to Review 2

Week	Dates	My milestone	Deliverable / check
1	25–31 Aug	Define CRITICAL / NORMAL classes and thresholds with the team; design the gateway threads (receiver + forwarder)	Design note and agreed record format
2	1–7 Sep	Implement the gateway: UDP receive, threshold classification, priority queue	Gateway forwards packets in priority order
3	8–14 Sep	Alert mode (throttle normal traffic during an alert); integration with sensors and server; literature study for the ML classifier	Gateway shown in the Review 2 demo

R1 Video Plan

Segment	What I show
Problem and evidence (1.5 min)	Face + screen; IMD / WMO evidence and the problem statement
Literature gap (1.5 min)	Face + screen; literature table and the gap
Proposed solution and feasibility (2 min)	Face + screen; block diagram and feasibility table
My role and timeline (1 min)	Face + screen; role table and weekly plan

References

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